

NETWORK RECONNAISSANCE AND SECURITY ASSESSMENT

Minor Project Report

Submitted as Part of the Industrial Training Program

Submitted By

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Cyber Security

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CHAPTER 1

Introduction

1.1. Introduction

Network reconnaissance is the process of collecting information about devices, services, operating systems, and communication within a network. It is the first phase of a security assessment and helps security professionals understand the network structure, identify active hosts, detect exposed services, and evaluate potential security risks before conducting further security testing.

In this project, a virtual laboratory environment was created using Oracle VirtualBox. Kali Linux was configured as the scanning machine, while Ubuntu Server acted as the target system. Various reconnaissance and analysis techniques were performed using Netdiscover, Nmap, and Wireshark to discover hosts, identify open ports and services, perform operating system fingerprinting, and capture network traffic. The collected information was analyzed to assess the overall security posture of the lab network and provide security recommendations.

1.2. Project Objective

The objective of this project is to assess the security posture of a virtual laboratory network by performing network reconnaissance, host discovery, port and service enumeration, operating system fingerprinting, and packet analysis using industry-standard cybersecurity tools.

1.3. Scope

The scope of this project includes:

- Host discovery and network mapping.
- Port and service enumeration.
- Operating system fingerprinting.
- Packet capture and analysis using Wireshark.
- Identification of exposed services and potential security risks.
- Security recommendations based on the findings.

1.4. Project Environment

Table 1.1 Project Environment

Component	Description
Host Machine	Windows 10
Virtualization Software	Oracle VirtualBox
Scanner Machine	Kali Linux
Target Machine	Ubuntu Server
Host-Only Network	192.168.56.0/24

Component	Description
NAT Network	Internet Connectivity
Network Assessment Tools	Netdiscover, Nmap, Wireshark

1.5. Methodology

The security assessment followed a structured methodology consisting of five phases:

1. **Lab Setup:** Configure Kali Linux and Ubuntu Server virtual machines using Host-Only and NAT networking.
2. **Host Discovery:** Identify active hosts on the virtual network using Netdiscover and Nmap.
3. **Port and Service Enumeration:** Detect open ports and running services using Nmap.
4. **Operating System Fingerprinting:** Identify the target operating system using Nmap's OS detection.
5. **Packet Analysis:** Capture and analyze ARP, ICMP, TCP, and SSH packets using Wireshark to observe network communication.

1.6. Network Topology

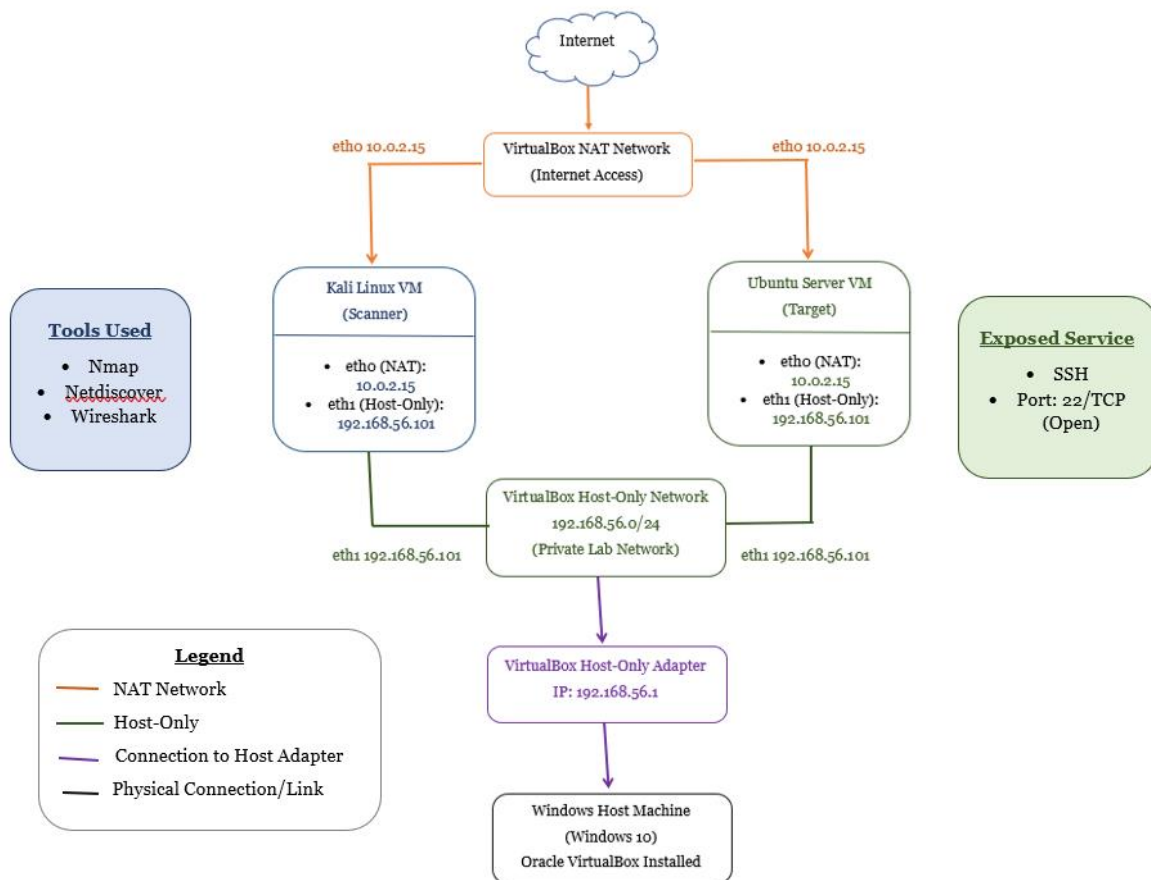


Figure 1.1 Virtual Lab Network Topology used for Network Reconnaissance and Security Assessment.

CHAPTER 2

Tools Used

2.1. Introduction

A variety of cybersecurity and network analysis tools were used to perform the network reconnaissance and security assessment. Each tool served a specific purpose, ranging from creating the virtual lab environment to discovering hosts, identifying services, and analyzing network traffic. The following sections describe the tools used during the project.

2.2. Oracle VirtualBox

Purpose: Virtualization Platform

Oracle VirtualBox is a cross-platform virtualization software that allows multiple operating systems to run simultaneously on a single physical computer. In this project, VirtualBox was used to create an isolated virtual laboratory consisting of Kali Linux and Ubuntu Server virtual machines. The virtual environment provided a safe platform to perform network reconnaissance without affecting the host operating system or external networks.

Role in Project

- Created the virtual lab environment.
- Hosted Kali Linux and Ubuntu Server virtual machines.
- Configured NAT and Host-Only networking.
- Enabled isolated communication between virtual machines.

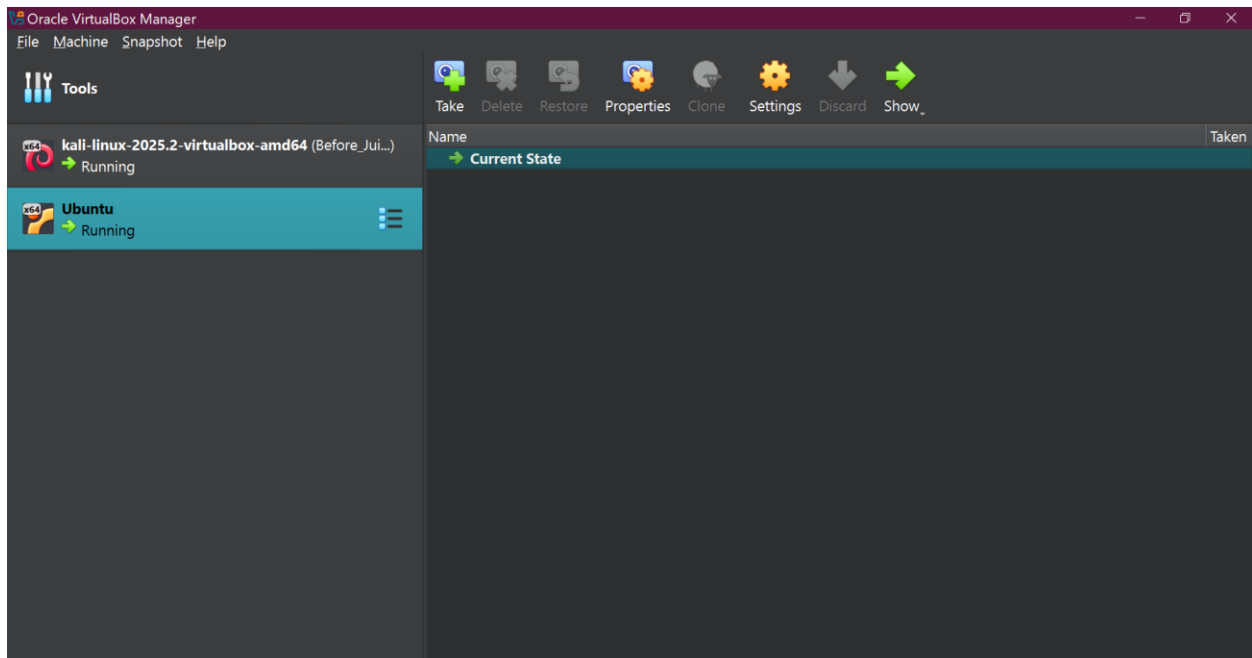


Figure 2.1 Oracle VirtualBox showing Kali Linux and Ubuntu Server virtual machines.

2.3. Kali Linux

Purpose: Security Assessment Platform

Kali Linux is a Debian-based Linux distribution designed specifically for cybersecurity, penetration testing, and digital forensics. It includes hundreds of pre-installed security tools used for network analysis, vulnerability assessment, and ethical hacking.

During this project, Kali Linux served as the scanning machine from which all reconnaissance activities were performed.

Role in Project

- Executed network scans.
- Performed host discovery.
- Conducted port and service enumeration.
- Captured network traffic using Wireshark.



Figure 2.2 Kali Linux virtual machine.

2.4. Ubuntu Server

Purpose: Target Machine

Ubuntu Server is a Linux server operating system widely used for hosting applications and network services. In this project, Ubuntu Server acted as the target machine against which reconnaissance activities were performed.

The SSH service was enabled on Ubuntu Server to demonstrate service detection and secure remote communication.

Role in Project

- Acted as the target system.

- Hosted the SSH service.
- Responded to reconnaissance requests.
- Generated network traffic for packet analysis.

```

Selecting previously unselected package ncurses-term.
Preparing to unpack ../2-ncurses-term_6.6+20251231-1_all.deb...
Unpacking ncurses-term (6.6+20251231-1)...
Selecting previously unselected package openssh-sftp-server.
Preparing to unpack ../3-openssh-sftp-server_1:9.10.2p1-2ubuntu3.5_amd64.deb...
Unpacking openssh-sftp-server (1:9.10.2p1-2ubuntu3.5)...
Selecting previously unselected package openssh-server.
Preparing to unpack ../4-openssh-server_1:9.10.2p1-2ubuntu3.5_amd64.deb...
Unpacking openssh-server (1:9.10.2p1-2ubuntu3.5)...
Selecting previously unselected package ssh-import-id.
Preparing to unpack ../5-ssh-import-id_5.11-0ubuntu4_all.deb...
Unpacking ssh-import-id (5.11-0ubuntu4)...
Setting up openssh-client (1:9.10.2p1-2ubuntu3.5)...
Setting up ssh-import-id (5.11-0ubuntu4)...
Setting up liburip0:amd64 (7.6.q-36build2)...
Setting up ncurses-term (6.6+20251231-1)...
Setting up openssh-sftp-server (1:9.10.2p1-2ubuntu3.5)...
Setting up openssh-server (1:9.10.2p1-2ubuntu3.5)...
Creating config file /etc/ssh/sshd_config with new version
Creating user 'sshd' (sshd user) with UID 983 and GID 65534.
Created symlink /etc/systemd/system/sockets.target.wants/ssh.socket → /usr/lib/systemd/system/ssh.socket'.
Created symlink /etc/systemd/system/ssh.service.requires/ssh.socket → /usr/lib/systemd/system/ssh.socket'.
Created symlink /etc/systemd/system/ssh.service.wants/sshd-keygen.service → /usr/lib/systemd/system/sshd-keygen.service'.
Created symlink /etc/systemd/system/sshd.service.wants/sshd-keygen.service → /usr/lib/systemd/system/sshd-keygen.service'.
Created symlink /etc/systemd/system/sshd.socket.wants/sshd-keygen.service → /usr/lib/systemd/system/sshd-keygen.service'.
Processing triggers for ufw (0.36.2-9build1)...
Processing triggers for man-db (2.13.1-1build1)...
Processing triggers for libc-bin (2.43-2ubuntu2)...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
vboxuser@Ubuntu:~$ sudo systemctl status ssh
* ssh.service - OpenSSH Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
   Active: inactive (dead)
TriggersBy: • ssh.socket
           Docs: man:sshd(8)
              man:sshd_config(5)
vboxuser@Ubuntu:~$ sudo systemctl start ssh
vboxuser@Ubuntu:~$

```

Figure 2.3 Ubuntu Server virtual machine.

2.5. Netdiscover

Purpose: Host Discovery

Netdiscover is an active and passive network reconnaissance tool used to identify live hosts within a local network using the Address Resolution Protocol (ARP).

During this project, Netdiscover was used to identify devices connected to the Host-Only network and determine their IP and MAC addresses.

Role in Project

- Discovered active hosts.
- Identified IP addresses.
- Displayed corresponding MAC addresses.
- Assisted in network mapping.

Command Used

`sudo netdiscover -i eth1`

Session Actions Edit View Help

Currently scanning: 172.27.0.0/16 | Screen View: Unique Hosts

6 Captured ARP Req/Rep packets, from 3 hosts. Total size: 360

IP	At MAC Address	Count	Len	MAC Vendor / Hostname
192.168.56.1	0a:00:27:00:00:0b	1	60	Unknown vendor
192.168.56.100	08:00:27:0e:52:2c	4	240	PCS Systemtechnik GmbH
192.168.56.104	08:00:27:83:ef:e9	1	60	PCS Systemtechnik GmbH

Figure 2.4 Host discovery using Netdiscover.

2.6. Nmap (Network Mapper)

Purpose: Network Scanning and Enumeration

Nmap is one of the most widely used open-source network scanning tools. It is capable of discovering hosts, scanning ports, detecting running services, identifying service versions, and performing operating system fingerprinting.

In this project, Nmap was used throughout multiple phases of the assessment.

Role in Project

- Host Discovery
- Port Enumeration
- Service Detection
- OS Fingerprinting

Commands Used

```
nmap -sn 192.168.56.0/24
```

```
nmap 192.168.56.104
```

```
nmap -sV 192.168.56.104
```

```
sudo nmap -O 192.168.56.104
```

```
(kali㉿kali)-[~]  
$ nmap -sn 192.168.56.0/24  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-04 02:26 -0400  
Nmap scan report for 192.168.56.1  
Host is up (0.00058s latency).  
MAC Address: 0A:00:27:00:00:0B (Unknown)  
Nmap scan report for 192.168.56.100  
Host is up (0.00061s latency).  
MAC Address: 08:00:27:0E:52:2C (Oracle VirtualBox virtual NIC)  
Nmap scan report for 192.168.56.104  
Host is up (0.00093s latency).  
MAC Address: 08:00:27:83:EF:E9 (Oracle VirtualBox virtual NIC)  
Nmap scan report for 192.168.56.101  
Host is up.  
Nmap done: 256 IP addresses (4 hosts up) scanned in 3.63 seconds
```

```
(kali㉿kali)-[~]  
$ nmap 192.168.56.104  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-04 02:26 -0400  
Nmap scan report for 192.168.56.104  
Host is up (0.00088s latency).  
Not shown: 999 closed tcp ports (reset)  
PORT      STATE SERVICE  
22/tcp    open  ssh  
MAC Address: 08:00:27:83:EF:E9 (Oracle VirtualBox virtual NIC)  
  
Nmap done: 1 IP address (1 host up) scanned in 1.96 seconds
```

```
(kali㉿kali)-[~]  
$ nmap -sV 192.168.56.104  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-04 02:30 -0400  
Nmap scan report for 192.168.56.104  
Host is up (0.0097s latency).  
Not shown: 999 closed tcp ports (reset)  
PORT      STATE SERVICE VERSION  
22/tcp    open  ssh      OpenSSH 10.2p1 Ubuntu 2ubuntu3.5 (Ubuntu Linux; protocol 2.0)  
MAC Address: 08:00:27:83:EF:E9 (Oracle VirtualBox virtual NIC)  
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 1.82 seconds
```

```
Nmap done: 1 IP address (1 host up) scanned in 11.02 seconds

(kali@kali)-[~]
$ sudo nmap -O 192.168.56.104
[sudo] password for kali:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-04 02:32 -0400
Nmap scan report for 192.168.56.104
Host is up (0.0015s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
MAC Address: 08:00:27:83:EF:E9 (Oracle VirtualBox virtual NIC)
No exact OS matches for host (If you know what OS is running on it, see https://nmap.org/submit/ ).
TCP/IP fingerprint:
OS:SCAN(V=7.99%E=4%D=8/4%OT=22%CT=1%CU=39214%PV=Y%DS=1%DC=D%G=Y%M=080027%TM
OS:=6A718796%P=x86_64-pc-linux-gnu)SEQ(SP=100%GCD=1%ISR=10A%TI=Z%CI=Z%TS=22
OS:)%SEQ(SP=101%GCD=1%ISR=10E%TI=Z%CI=Z%TS=21)SEQ(SP=102%GCD=1%ISR=10D%TI=Z%
OS:CI=Z%II=I%TS=22)SEQ(SP=102%GCD=2%ISR=109%TI=Z%CI=Z%II=I%TS=21)SEQ(SP=107
OS:%GCD=1%ISR=10E%TI=Z%CI=Z%II=I%TS=21)OPS(O1=M5B4ST11NW7%O2=M5B4ST11NW7%O3
OS:=M5B4NNT11NW7%O4=M5B4ST11NW7%O5=M5B4ST11NW7%O6=M5B4ST11)WIN(W1=FE88%W2=F
OS:E88%W3=FE88%W4=FE88%W5=FE88%W6=FE88)ECN(R=Y%DF=Y%T=40%W=FAF0%O=M5B4NNSNW
OS:7%CC=Y%Q=)T1(R=Y%DF=Y%T=40%S=0%A=S+%F=AS%RD=0%Q=)T2(R=N)T3(R=N)T4(R=Y%DF
OS:=Y%T=40%W=0%S=A%A=Z%F=R%O=0%RD=0%Q=)T5(R=Y%DF=Y%T=40%W=0%S=Z%A=S+%F=AR%O=
OS:%RD=0%Q=)T6(R=Y%DF=Y%T=40%W=0%S=A%A=Z%F=R%O=0%RD=0%Q=)T7(R=Y%DF=Y%T=40%W=
OS:0%S=Z%A=S+%F=AR%O=0%RD=0%Q=)U1(R=Y%DF=N%T=40%IPL=164%UN=0%RIPL=G%RID=G%RI
OS:PCK=G%RUCK=G%RUD=G)IE(R=Y%DFI=N%T=40%CD=S)

Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 12.28 seconds
```

Figure 2.5 Nmap scan results.

2.7. Wireshark

Purpose: Packet Analysis

Wireshark is an open-source network protocol analyzer used to capture and inspect packets travelling across a network. It enables detailed analysis of communication between devices at different protocol layers.

During this project, Wireshark was used to capture and analyze ARP, ICMP, TCP, and SSH packets exchanged between Kali Linux and Ubuntu Server.

Role in Project

- Captured live network traffic.
- Analyzed ICMP communication.
- Observed ARP requests and replies.
- Examined TCP packets during port scanning.
- Verified encrypted SSH communication.

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Capturing from eth1

icmp

No.	Time	Source	Destination	Protocol	Length	Info
5	11.923658182	192.168.56.101	192.168.56.184	ICMP	98	Echo (ping) request id=0xc12b, seq=17256, ttl=64 (reply in 6)
6	11.923658182	192.168.56.101	192.168.56.184	ICMP	98	Echo (ping) request id=0xc12b, seq=17256, ttl=64 (reply in 6)
7	13.315730919	192.168.56.101	192.168.56.184	ICMP	98	Echo (ping) request id=0xc12b, seq=2/512, ttl=64 (reply in 8)
8	13.316849395	192.168.56.184	192.168.56.101	ICMP	98	Echo (ping) reply id=0xc12b, seq=2/512, ttl=64 (request in 7)
9	14.316491149	192.168.56.101	192.168.56.184	ICMP	98	Echo (ping) request id=0xc12b, seq=3/768, ttl=64 (reply in 10)
10	14.317618393	192.168.56.184	192.168.56.101	ICMP	98	Echo (ping) reply id=0xc12b, seq=3/768, ttl=64 (request in 9)
11	15.327383115	192.168.56.101	192.168.56.184	ICMP	98	Echo (ping) request id=0xc12b, seq=4/1824, ttl=64 (reply in 12)
12	15.328625931	192.168.56.184	192.168.56.101	ICMP	98	Echo (ping) reply id=0xc12b, seq=4/1824, ttl=64 (request in 11)

Frame 6: Packet, 98 bytes on wire (784 bits), 98 bytes capture
 Section number: 1
 Interface id: 0 (eth1)
 Encapsulation type: Ethernet (1)
 Arrival Time: Aug 4, 2020 02:54:26.247404146 EDT
 UTC Arrival Time: Aug 4, 2020 06:54:26.247404146 UTC
 Epoch Arrival Time: 1785826466.247404146
 [Time shift for this packet: 0.000000000 seconds]
 [Time delta from previous captured frame: 1.435207 millisecc]
 [Time delta from previous displayed frame: 1.435207 millisecc]
 [Time since reference or first frame: 11.925993309 seconds]
 Frame Number: 6
 Frame Length: 98 bytes (784 bits)
 Capture Length: 98 bytes (784 bits)
 [Frame is marked: False]
 [Frame is ignored: False]
 [Protocols in frame: ethertype:ip:icmp:data]
 Character encoding: ASCII (0)

Internet Control Message Protocol: Protocol

Packets: 16 - Displayed: 8 (50.0%)

Profile: Default

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Capturing from eth1

arp

No.	Time	Source	Destination	Protocol	Length	Info
3	11.921843784	PCSSystemtec_74:ae:..	Broadcast	ARP	42	who has 192.168.56.184? Tell 192.168.56.101
4	11.921843784	PCSSystemtec_83:ef:..	PCSSystemtec_74:ae:..	ARP	42	who has 192.168.56.184? Tell 192.168.56.101
13	17.420240103	PCSSystemtec_83:ef:..	PCSSystemtec_74:ae:..	ARP	60	who has 192.168.56.101? Tell 192.168.56.184
14	17.420258861	PCSSystemtec_74:ae:..	PCSSystemtec_83:ef:..	ARP	42	192.168.56.101 is at 08:00:27:74:ae:88
26	164.238469144	PCSSystemtec_74:ae:..	PCSSystemtec_83:ef:..	ARP	42	who has 192.168.56.184? Tell 192.168.56.101
27	164.239494661	PCSSystemtec_83:ef:..	PCSSystemtec_74:ae:..	ARP	60	192.168.56.184 is at 08:00:27:83:ef:e9
28	164.437392509	PCSSystemtec_83:ef:..	PCSSystemtec_74:ae:..	ARP	60	who has 192.168.56.101? Tell 192.168.56.184
29	164.437373871	PCSSystemtec_74:ae:..	PCSSystemtec_83:ef:..	ARP	42	192.168.56.101 is at 08:00:27:74:ae:88

Frame 4: Packet, 60 bytes on wire (480 bits), 60 bytes capture
 Section number: 1
 Interface id: 0 (eth1)
 Encapsulation type: Ethernet (1)
 Arrival Time: Aug 4, 2020 02:54:26.245953925 EDT
 UTC Arrival Time: Aug 4, 2020 06:54:26.245953925 UTC
 Epoch Arrival Time: 1785826466.245953925
 [Time shift for this packet: 0.000000000 seconds]
 [Time delta from previous captured frame: 1.799394 millisecc]
 [Time delta from previous displayed frame: 1.799394 millisecc]
 [Time since reference or first frame: 11.923643088 seconds]
 Frame Number: 4
 Frame Length: 60 bytes (480 bits)
 Capture Length: 60 bytes (480 bits)
 [Frame is marked: False]
 [Frame is ignored: False]
 [Protocols in frame: ethertype:arp]
 Character encoding: ASCII (0)

Address Resolution Protocol: Protocol

Packets: 29 - Displayed: 8 (27.6%)

Profile: Default

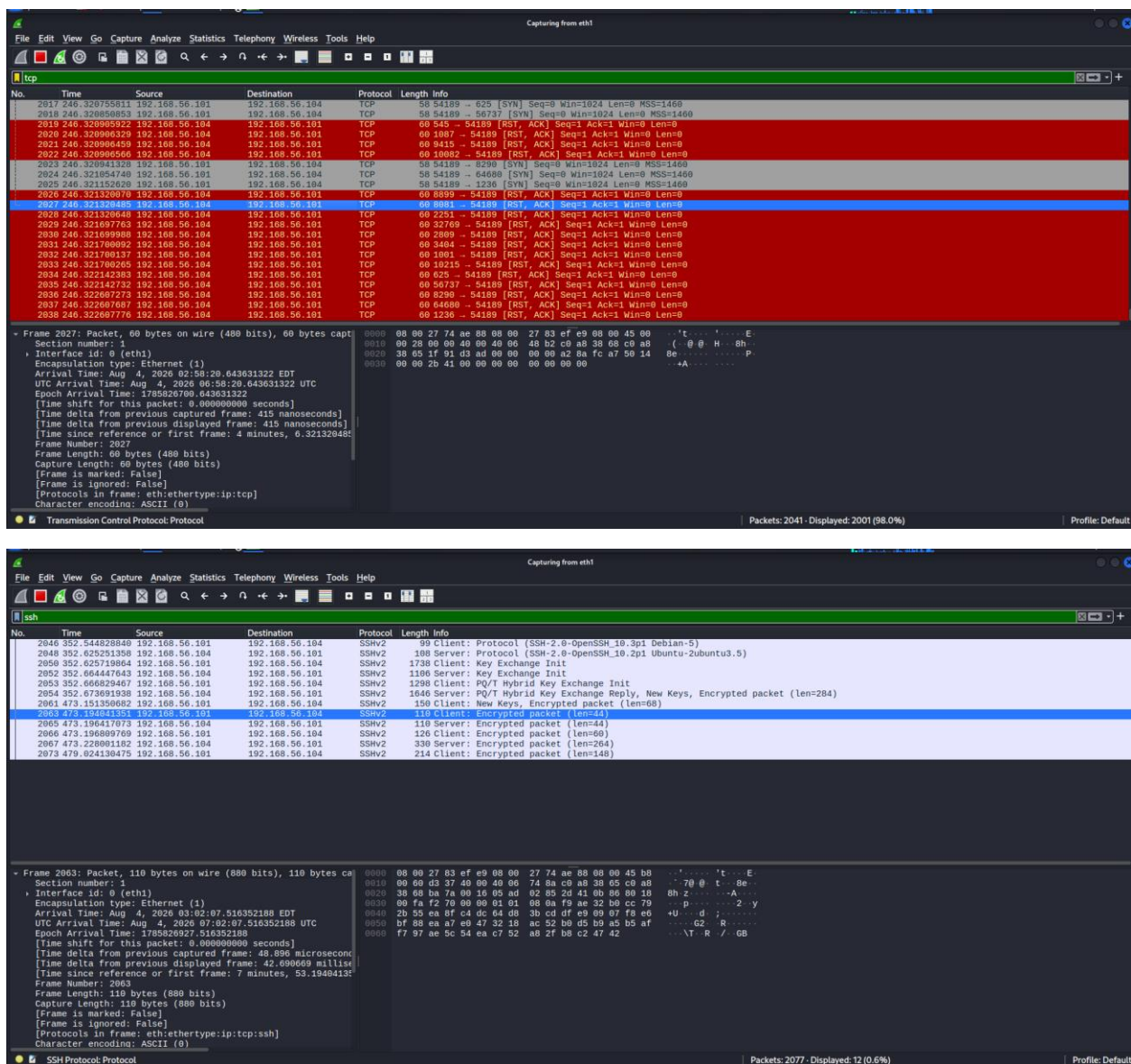


Figure 2.6 Wireshark packet capture.

2.8. Summary

The selected tools collectively provided a complete environment for performing network reconnaissance and security assessment. Oracle VirtualBox created the virtual lab, Kali Linux served as the scanning platform, Ubuntu Server acted as the target system, Netdiscover identified active hosts, Nmap performed reconnaissance and enumeration, and Wireshark analyzed the captured network traffic. Together, these tools enabled a structured assessment of the lab network.

CHAPTER 3

Lab Setup

3.1. Introduction

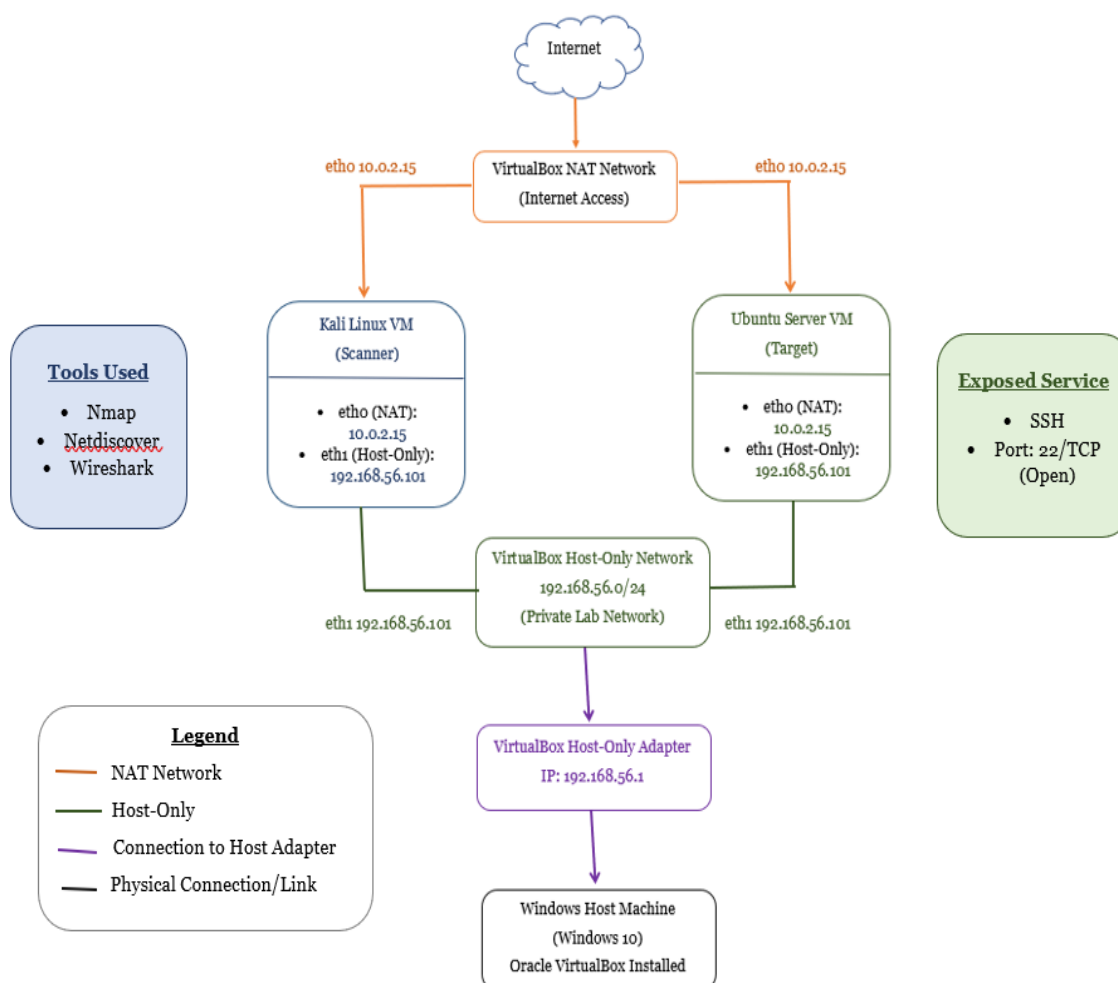
Before performing the network reconnaissance and security assessment, an isolated virtual laboratory was created using Oracle VirtualBox. The lab consisted of two virtual machines: Kali Linux, which acted as the scanning machine, and Ubuntu Server, which served as the target machine. A combination of Host-Only and NAT networking was used to provide secure communication between the virtual machines while maintaining Internet connectivity for software installation and updates.

3.2. Lab Architecture

The virtual lab consisted of the following components:

Table 3.1 Virtual Laboratory Components

Component	Configuration
Host Machine	Windows 10
Virtualization Platform	Oracle VirtualBox
Scanner Machine	Kali Linux
Target Machine	Ubuntu Server
Host-Only Network	192.168.56.0/24
NAT Network	Internet Connectivity



3.3. Virtual Machine Configuration

Kali Linux (Scanner)

Kali Linux was configured as the scanning and analysis machine. It contained the required security tools, including Netdiscover, Nmap, and Wireshark, which were used throughout the assessment.

Table 2.2 Kali Linux Virtual Machine Configuration

Parameter	Value
Operating System	Kali Linux
Role	Scanner
Adapter 1	NAT
Adapter 2	Host-Only Adapter
Host-Only IP Address	192.168.56.101

Ubuntu Server (Target)

Ubuntu Server was configured as the target machine for reconnaissance activities. The OpenSSH service was installed and enabled to demonstrate port scanning and service detection.

Table 3.3 Ubuntu Server Virtual Machine Configuration

Parameter	Value
Operating System	Ubuntu Server
Role	Target
Adapter 1	NAT
Adapter 2	Host-Only Adapter
Host-Only IP Address	192.168.56.104

3.4. Network Configuration

Both virtual machines were connected using two different network adapters.

NAT Network

The NAT adapter provided Internet connectivity to both virtual machines. It was primarily used for downloading software packages and installing required services such as OpenSSH.

Host-Only Network

The Host-Only network created an isolated communication channel between Kali Linux and Ubuntu Server. All reconnaissance activities, including host discovery, port scanning, service detection, and packet capture, were performed over this network.

3.5. IP Address Verification

The IP addresses of both virtual machines were verified using the following command:

```
ip addr
```

Results

Table 3.4 Verified IP Address Configuration

Virtual Machine	Host-Only IP Address
Kali Linux	192.168.56.101
Ubuntu Server	192.168.56.104

```

(kali@kali)-[~]
$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:d1:f8:5d brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 86218sec preferred_lft 86218sec
    inet6 fd17:625c:f037:2:32a2:2cd7:47d7:ff98/64 scope global dynamic noprefixroute
        valid_lft 86220sec preferred_lft 14220sec
    inet6 fe80::6f15:aee6:9143:cc7d/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:74:ae:88 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.101/24 brd 192.168.56.255 scope global dynamic noprefixroute eth1
        valid_lft 418sec preferred_lft 418sec
    inet6 fe80::90c9:346f:5a4c:a31/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
4: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
    link/ether d6:89:87:ef:1c:cb brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
        valid_lft forever preferred_lft forever

```

Figure 3.1 IP address configuration of Kali Linux.

```

vboxuser@Ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:32:4d:a0 brd ff:ff:ff:ff:ff:ff
    altname enx080027324da0
    inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
        valid_lft 86283sec preferred_lft 86283sec
    inet6 fd17:625c:f037:2:a00:27ff:fe32:4da0/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 86284sec preferred_lft 14284sec
    inet6 fe80::a00:27ff:fe32:4da0/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:83:ef:e9 brd ff:ff:ff:ff:ff:ff
    altname enx08002783efe9
    inet 192.168.56.104/24 metric 1024 brd 192.168.56.255 scope global dynamic enp0s8
        valid_lft 483sec preferred_lft 483sec
    inet6 fe80::a00:27ff:fe83:efe9/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
vboxuser@Ubuntu:~$

```

Figure 3.2 IP address configuration of Ubuntu Server.

Connectivity Verification

Before performing reconnaissance, connectivity between both virtual machines was verified using the ICMP Ping utility.

Command

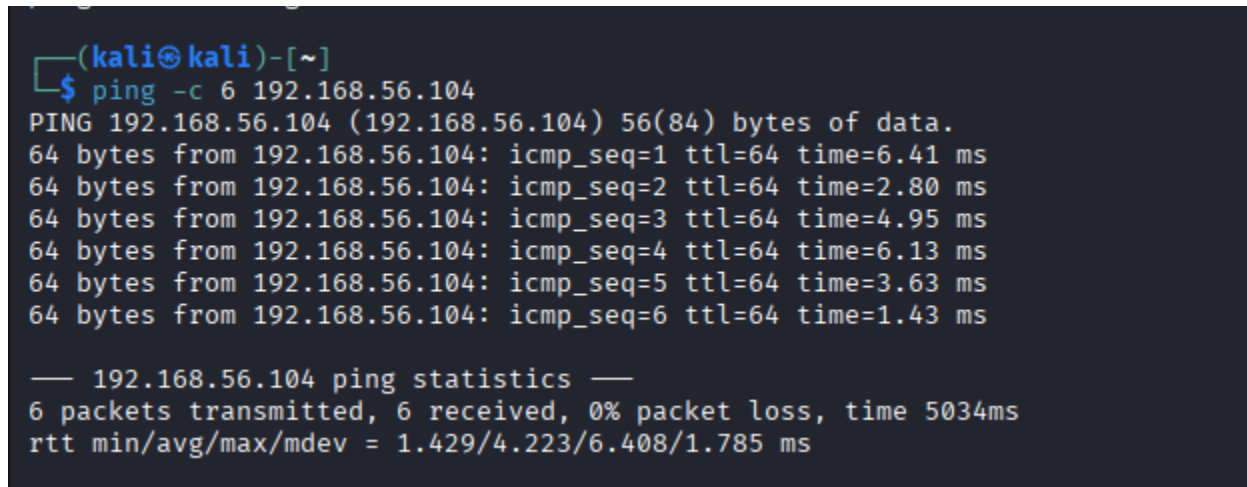
```
ping -c 4 192.168.56.104
```

Observation

The Ubuntu Server responded successfully to all four ICMP Echo Requests with no packet loss.

Result

Successful communication confirmed that both virtual machines were connected correctly through the Host-Only network and were ready for security assessment.



```
(kali㉿kali)-[~]
$ ping -c 6 192.168.56.104
PING 192.168.56.104 (192.168.56.104) 56(84) bytes of data.
64 bytes from 192.168.56.104: icmp_seq=1 ttl=64 time=6.41 ms
64 bytes from 192.168.56.104: icmp_seq=2 ttl=64 time=2.80 ms
64 bytes from 192.168.56.104: icmp_seq=3 ttl=64 time=4.95 ms
64 bytes from 192.168.56.104: icmp_seq=4 ttl=64 time=6.13 ms
64 bytes from 192.168.56.104: icmp_seq=5 ttl=64 time=3.63 ms
64 bytes from 192.168.56.104: icmp_seq=6 ttl=64 time=1.43 ms

— 192.168.56.104 ping statistics —
6 packets transmitted, 6 received, 0% packet loss, time 5034ms
rtt min/avg/max/mdev = 1.429/4.223/6.408/1.785 ms
```

Figure 3.3 Successful ICMP Ping from Kali Linux to Ubuntu Server.

3.6. SSH Service Configuration

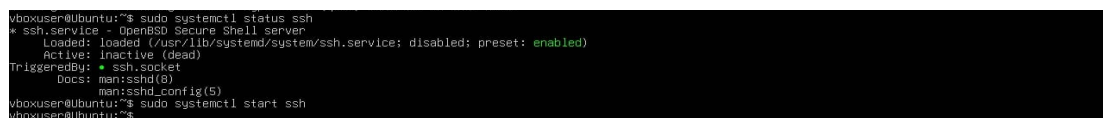
To perform service enumeration, the OpenSSH Server package was installed and configured on the Ubuntu Server.

Commands Used

```
sudo apt update
sudo apt install openssh-server
sudo systemctl enable ssh
sudo systemctl start ssh
sudo systemctl status ssh
```

Observation

The SSH service was successfully installed, enabled, and started. Verification using the `systemctl status ssh` command confirmed that the service was active and running.



```
vboxuser@Ubuntu:~$ sudo systemctl status ssh
* ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
   Active: inactive (dead)
   TriggeredBy: ● ssh.socket
   Docs: man:sshd(8)
          man:sshd_config(5)
vboxuser@Ubuntu:~$ sudo systemctl start ssh
vboxuser@Ubuntu:~$
```

Figure 3.4 SSH service running on Ubuntu Server.

3.7. Chapter Summary

A secure and isolated virtual laboratory was successfully configured using Oracle VirtualBox. Kali Linux and Ubuntu Server were connected through both NAT and Host-Only networks, enabling

Internet connectivity and private communication. IP configuration, connectivity testing, and SSH service setup were completed successfully, providing a stable environment for the subsequent reconnaissance and security assessment activities.

CHAPTER 4

Host Discovery and Network Mapping

4.1. Introduction

Host discovery is the first step in network reconnaissance. It is used to identify active devices connected to a network before performing detailed security assessments. In this project, host discovery was performed using **Netdiscover** and **Nmap** to identify live hosts within the Host-Only network and to create an understanding of the network topology.

4.2. Objective

The objective of this phase was to:

- Discover active hosts on the Host-Only network.
- Identify IP and MAC addresses of connected devices.
- Verify communication between the scanner and target systems.
- Create a basic map of the virtual lab network.

4.3. Host Discovery using Netdiscover

Tool Used

Netdiscover

Netdiscover was used to perform Address Resolution Protocol (ARP)-based host discovery within the Host-Only network. Since the Kali Linux virtual machine was configured with both NAT and Host-Only adapters, the Host-Only interface (eth1) was explicitly specified to ensure the scan was performed on the correct network.

Command Used

```
sudo netdiscover -i eth1
```

Observation

The scan successfully detected active devices connected to the Host-Only network. The identified hosts included the VirtualBox Host-Only Adapter, VirtualBox DHCP Server, and the Ubuntu Server target machine.

Scan Results

Table 4.1 Netdiscover Host Discovery Results

IP Address	Device	Description
192.168.56.1	VirtualBox Host-Only Adapter	Gateway/Host interface
192.168.56.100	VirtualBox DHCP Server	Assigns IP addresses
192.168.56.104	Ubuntu Server	Target machine

Session Actions Edit View Help

Currently scanning: 172.27.0.0/16 | Screen View: Unique Hosts

6 Captured ARP Req/Rep packets, from 3 hosts. Total size: 360

IP	At MAC Address	Count	Len	MAC Vendor / Hostname
192.168.56.1	0a:00:27:00:00:0b	1	60	Unknown vendor
192.168.56.100	08:00:27:0e:52:2c	4	240	PCS Systemtechnik GmbH
192.168.56.104	08:00:27:83:ef:e9	1	60	PCS Systemtechnik GmbH

Figure 7 Host discovery using Netdiscover.

Inference

The Netdiscover scan confirmed that the target Ubuntu Server was reachable on the Host-Only network. The identified devices matched the expected virtual lab configuration.

4.4. Host Discovery using Nmap

After identifying hosts with Netdiscover, **Nmap** was used to verify all active systems on the network.

Command Used

```
nmap -sn 192.168.56.0/24
```

Observation

The scan identified four active hosts within the virtual network.

Scan Results

Table 4.2 Nmap Host Discovery Results

IP Address	Device
192.168.56.1	VirtualBox Host-Only Adapter
192.168.56.100	VirtualBox DHCP Server
192.168.56.101	Kali Linux
192.168.56.104	Ubuntu Server

```
(kali㉿kali)-[~]
$ nmap -sn 192.168.56.0/24
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-04 02:26 -0400
Nmap scan report for 192.168.56.1
Host is up (0.00058s latency).
MAC Address: 0A:00:27:00:00:0B (Unknown)
Nmap scan report for 192.168.56.100
Host is up (0.00061s latency).
MAC Address: 08:00:27:0E:52:2C (Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.56.104
Host is up (0.00093s latency).
MAC Address: 08:00:27:83:EF:E9 (Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.56.101
Host is up.
Nmap done: 256 IP addresses (4 hosts up) scanned in 3.63 seconds
```

Figure 8 Nmap Host Discovery Scan.

Comparison of Netdiscover and Nmap

Table 4.3 Comparison of Netdiscover and Nmap

Feature	Netdiscover	Nmap
Discovery Method	ARP	ICMP/ARP/Ping
Detects Local Hosts	Yes	Yes
Shows MAC Addresses	Yes	No (in basic scan)
Shows Scanner Machine	Usually No	Yes
Best Use	Local Network Discovery	General Host Discovery

Analysis

Both Netdiscover and Nmap successfully identified the active hosts within the virtual network.

Netdiscover provided detailed ARP-level information, including MAC addresses, while Nmap confirmed the availability of all active hosts through a ping scan. The results from both tools were consistent with the configured virtual laboratory.

4.5. Network Mapping

Based on the host discovery results, the network consisted of four active devices:

Table 4.4 Network Mapping Summary

Device	IP Address	Role
Windows Host	192.168.56.1	Host System
VirtualBox DHCP Server	192.168.56.100	DHCP Service
Kali Linux	192.168.56.101	Scanner
Ubuntu Server	192.168.56.104	Target

The network mapping confirmed that all expected devices were active and reachable before proceeding with detailed security assessment activities.

4.6. Chapter Summary

Host discovery was successfully performed using both Netdiscover and Nmap. The active hosts detected matched the configured virtual lab environment, confirming proper network connectivity. These results established a reliable foundation for the subsequent phases of port scanning, service enumeration, and operating system fingerprinting.

CHAPTER 5

Port and Service Enumeration

5.1. Introduction

Port and service enumeration is a critical phase of network reconnaissance. It helps identify open ports, determine the services running on those ports, and understand the potential attack surface of a target system. The information gathered during this phase assists security professionals in identifying accessible services that may require further security evaluation.

In this project, Nmap was used to identify open TCP ports and detect the versions of the services running on the Ubuntu Server.

5.2. Objective

The objectives of this phase were to:

- Identify open TCP ports on the target system.
- Detect the services running on the identified ports.
- Determine the service version.
- Evaluate the potential security exposure of the detected services.

5.3. Port Enumeration using Nmap

Tool Used

Nmap (Network Mapper)

Port enumeration was performed to identify the open TCP ports available on the Ubuntu Server.

Command Used

```
nmap 192.168.56.104
```

Observation

The scan identified **one open TCP port** on the target system.

Scan Results

Table 5.1Port Enumeration Results

Port	Protocol	State	Service
22	TCP	Open	SSH

The remaining **999 commonly scanned TCP ports were found to be closed**, indicating that only the SSH service was exposed.

```
(kali㉿kali)-[~]  
$ nmap 192.168.56.104  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-04 02:26 -0400  
Nmap scan report for 192.168.56.104  
Host is up (0.00088s latency).  
Not shown: 999 closed tcp ports (reset)  
PORT      STATE SERVICE  
22/tcp    open  ssh  
MAC Address: 08:00:27:83:EF:E9 (Oracle VirtualBox virtual NIC)  
  
Nmap done: 1 IP address (1 host up) scanned in 1.96 seconds
```

Figure 5.1 Nmap Port Scan Results.

Analysis

The scan results indicate that the Ubuntu Server exposes only the SSH service on Port 22. All other commonly scanned TCP ports were closed, resulting in a relatively small attack surface. Restricting unnecessary open ports is considered a good security practice because it reduces the number of potential entry points available to an attacker.

5.4. Service Enumeration using Nmap

After identifying the open port, service version detection was performed.

Command Used

```
nmap -sV 192.168.56.104
```

Observation

Nmap successfully identified the service running on Port 22.

Service Details

Table 5.2 Service Enumeration Results

Port	Service	Version
22/TCP	SSH	OpenSSH 10.2p1 (Ubuntu)

Additional information obtained during the scan indicated that the SSH service was using **SSH Protocol Version 2.0**, which is the recommended and secure version of the protocol.

```
(kali@kali)-[~]
$ nmap -sV 192.168.56.104
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-04 02:30 -0400
Nmap scan report for 192.168.56.104
Host is up (0.0097s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 10.2p1 Ubuntu 2ubuntu3.5 (Ubuntu Linux; protocol 2.0)
MAC Address: 08:00:27:83:EF:E9 (Oracle VirtualBox virtual NIC)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 1.82 seconds
```

Figure 5.2 Service Version Detection using Nmap.

Analysis

Service enumeration revealed that the target system was running **OpenSSH 10.2p1**. Identifying the exact software version is an important step in security assessments because it enables administrators and security analysts to compare the software against publicly known vulnerabilities and ensure that the service is up to date.

The use of **SSH Protocol Version 2.0** indicates that the server is configured to use the modern and secure version of the SSH protocol.

5.5. Findings

The following findings were identified during port and service enumeration:

Table 5.3 Port and Service Findings

Finding	Description
Open Port	TCP Port 22
Running Service	Secure Shell (SSH)
Service Version	OpenSSH 10.2p1
Protocol Version	SSH Protocol 2.0
Closed Ports	999 TCP ports

5.6. Security Assessment

Although the SSH service is intended for secure remote administration, any publicly accessible service can become a target for malicious activities if not properly configured.

Potential risks associated with an exposed SSH service include:

- Brute-force password attacks.
- Unauthorized login attempts.
- Exploitation of outdated SSH software.

- Weak authentication mechanisms.

However, no evidence of misconfiguration or additional exposed services was observed during this assessment.

5.7. Chapter Summary

Port enumeration identified a single open TCP port (22) on the Ubuntu Server. Service enumeration confirmed that the system was running OpenSSH 10.2p1 using SSH Protocol 2.0. The limited number of exposed services indicates a relatively small attack surface, although proper SSH security practices should continue to be followed.

Key Observation:

The Ubuntu Server exposed only the SSH service on Port 22, while the remaining 999 commonly scanned TCP ports were closed. This limited exposure reduces the attack surface and reflects a more secure network configuration.

CHAPTER 6

Operating System Fingerprinting

6.1. Introduction

Operating System (OS) fingerprinting is a reconnaissance technique used to identify the operating system running on a target machine. Knowledge of the operating system helps security professionals understand the target environment and determine whether specific vulnerabilities or security issues may exist. Nmap performs OS fingerprinting by analyzing the responses of the target system to specially crafted network packets and comparing them with its OS fingerprint database.

6.2. Objective

The objectives of this phase were to:

- Identify the operating system of the target machine.
- Verify whether the detected operating system matched the expected lab configuration.
- Collect additional system information for security assessment.

6.3. OS Fingerprinting using Nmap

Tool Used

Nmap (Network Mapper)

Command Used

```
sudo nmap -O 192.168.56.104
```

6.4. Observation

The OS fingerprinting scan detected that the target machine was running a Linux-based operating system. However, Nmap was unable to identify an exact operating system fingerprint and displayed the message:

"No exact OS matches for host."

The scan also reported:

- **Host Status:** Up
- **Network Distance:** 1 hop
- **Open Port:** 22/TCP (SSH)

6.5. Screenshot

```
Nmap done: 1 IP address (1 host up) scanned in 1.02 seconds

(kali@kali)-[~]
$ sudo nmap -O 192.168.56.104
[sudo] password for kali:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-04 02:32 -0400
Nmap scan report for 192.168.56.104
Host is up (0.0015s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
MAC Address: 08:00:27:83:EF:E9 (Oracle VirtualBox virtual NIC)
No exact OS matches for host (If you know what OS is running on it, see https://nmap.org/submit/ ).
TCP/IP fingerprint:
OS:SCAN(V=7.99%E=4%D=8/4%OT=22%CT=1%CU=39214%PV=Y%DS=1%DC=D%G=Y%M=080027%TM
OS:=6A718796%P=x86_64-pc-linux-gnu)SEQ(SP=100%GCD=1%ISR=10A%TI=Z%CI=Z%TS=22
OS: )SEQ(SP=101%GCD=1%ISR=10E%TI=Z%CI=Z%TS=21)SEQ(SP=102%GCD=1%ISR=10D%TI=Z%
OS:CI=Z%II=I%TS=22)SEQ(SP=102%GCD=2%ISR=109%TI=Z%CI=Z%II=I%TS=21)SEQ(SP=107
OS:%GCD=1%ISR=10E%TI=Z%CI=Z%II=I%TS=21)OPS(O1=M5B4ST11NW7%O2=M5B4ST11NW7%O3
OS:=M5B4NNT11NW7%O4=M5B4ST11NW7%O5=M5B4ST11NW7%O6=M5B4ST11)WIN(W1=FE88%W2=F
OS:E88%W3=FE88%W4=FE88%W5=FE88%W6=FE88)ECN(R=Y%DF=Y%T=40%W=FAF0%O=M5B4NNSNW
OS:7%CC=Y%Q=)T1(R=Y%DF=Y%T=40%S=0%A=S+%F=AS%RD=0%Q=)T2(R=N)T3(R=N)T4(R=Y%DF
OS:=Y%T=40%W=0%S=A%A=Z%F=R%O=%RD=0%Q=)T5(R=Y%DF=Y%T=40%W=0%S=Z%A=S+%F=AR%O=
OS:%RD=0%Q=)T6(R=Y%DF=Y%T=40%W=0%S=A%A=Z%F=R%O=%RD=0%Q=)T7(R=Y%DF=Y%T=40%W=
OS:0%S=Z%A=S+%F=AR%O=%RD=0%Q=)U1(R=Y%DF=N%T=40%IPL=164%UN=0%RIPL=G%RID=G%RI
OS:PCK=G%RUCK=G%RUD=G)IE(R=Y%DFI=N%T=40%CD=S)

Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 12.28 seconds
```

Figure 6.1 Operating System Fingerprinting using Nmap.

6.6. Analysis

Although Nmap did not identify an exact operating system match, the scan successfully confirmed that the target belonged to the Linux operating system family. Since the target machine in the laboratory was Ubuntu Server, the observed result was consistent with the expected configuration.

The inability to obtain an exact fingerprint does not necessarily indicate a problem. Several factors can affect OS detection, including:

- A limited number of open ports available for fingerprinting.
- The use of virtualization (Oracle VirtualBox).
- Minor differences between the observed fingerprint and Nmap's fingerprint database.
- Network configuration and firewall behavior.

Therefore, the result obtained during this assessment is considered normal for a virtual lab environment.

6.7. Findings

Table 6.1 Operating System Fingerprinting Results

Parameter	Result
Tool Used	Nmap
Command	sudo nmap -O 192.168.56.104

Parameter	Result
Target System	Ubuntu Server
Detected OS Family	Linux
Exact OS Match	Not Identified
Network Distance	1 Hop

6.8. Security Assessment

The successful identification of the Linux operating system family provides valuable information for security assessment. Knowledge of the operating system allows security professionals to:

- Identify operating system–specific vulnerabilities.
- Verify software compatibility.
- Select appropriate security assessment techniques.
- Recommend relevant security updates and patches.

No indication of operating system misconfiguration was identified during this phase of the assessment.

6.9. Chapter Summary

Operating system fingerprinting was successfully performed using Nmap. The scan identified the target as a Linux-based system but could not determine an exact operating system fingerprint. This result is common in virtualized environments and did not affect the overall security assessment. The findings confirmed that the target environment matched the expected Ubuntu Server configuration.

CHAPTER 7

Packet Analysis

7.1. Introduction

Packet analysis is the process of capturing and examining network traffic to understand how devices communicate over a network. It provides visibility into different network protocols, communication patterns, and data exchange between hosts. During a security assessment, packet analysis helps verify network connectivity, observe protocol behavior, and identify any unusual or unauthorized communication.

In this project, **Wireshark** was used to capture and analyze packets exchanged between the Kali Linux scanner and the Ubuntu Server target.

7.2. Objective

The objectives of packet analysis were to:

- Capture network packets generated during reconnaissance.
- Observe communication between the scanner and target machines.
- Analyze the behavior of common network protocols.
- Verify successful communication within the virtual lab environment.

7.3. Packet Capture Setup

Packet capture was performed on the **Host-Only network interface (eth1)** of the Kali Linux virtual machine. The Host-Only interface was selected because all communication between Kali Linux and Ubuntu Server occurred over the isolated lab network.

The following protocol filters were used during packet analysis:

- ARP
- ICMP
- TCP
- SSH

7.4. ARP Packet Analysis

Filter Used

arp

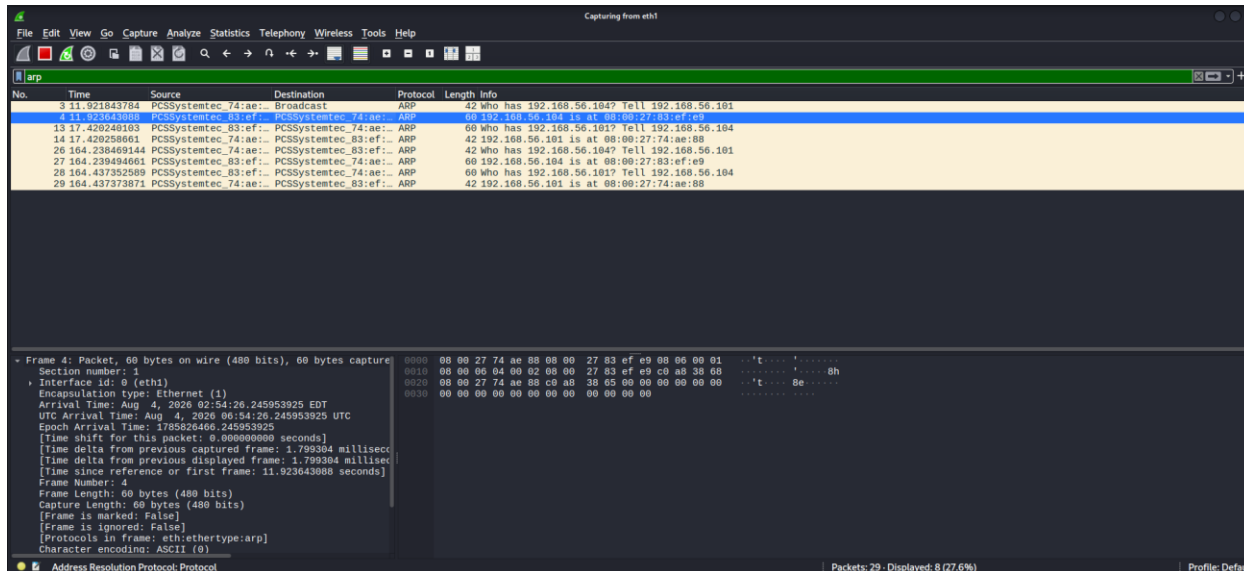
Observation

Wireshark captured ARP Request and ARP Reply packets exchanged between Kali Linux and Ubuntu Server.

The ARP Request asked for the MAC address associated with the Ubuntu Server IP address, while the ARP Reply provided the corresponding hardware address. This process enabled successful communication over the Ethernet network.

Analysis

The captured ARP packets confirmed that the scanner successfully resolved the target system's MAC address before initiating communication. ARP resolution is an essential process for communication within a local area network.



No.	Time	Source	Destination	Protocol	Length	Info
3	11.921843784	PCSSystemtec_74:ae	Broadcast	ARP	42	who has 192.168.56.184? Tell 192.168.56.181
4	11.923643888	PCSSystemtec_83:ef	PCSSystemtec_74:ae	ARP	60	192.168.56.184 is at 08:00:27:83:ef:e9
13	17.428248183	PCSSystemtec_83:ef	PCSSystemtec_74:ae	ARP	60	who has 192.168.56.181? Tell 192.168.56.184
14	17.428258861	PCSSystemtec_74:ae	PCSSystemtec_83:ef	ARP	42	192.168.56.181 is at 08:00:27:74:ae:88
26	164.238469144	PCSSystemtec_74:ae	PCSSystemtec_83:ef	ARP	42	who has 192.168.56.184? Tell 192.168.56.181
27	164.239494601	PCSSystemtec_83:ef	PCSSystemtec_74:ae	ARP	60	192.168.56.184 is at 08:00:27:83:ef:e9
28	164.437352589	PCSSystemtec_83:ef	PCSSystemtec_74:ae	ARP	60	who has 192.168.56.181? Tell 192.168.56.184
29	164.437373871	PCSSystemtec_74:ae	PCSSystemtec_83:ef	ARP	42	192.168.56.181 is at 08:00:27:74:ae:88

Section number: 1
Interface id: 0 (eth1)
Encapsulation type: Ethernet (1)
Arrival Time: Aug 4, 2026 02:54:26.245953925 EDT
UTC Arrival Time: Aug 4, 2026 06:54:26.245953925 UTC
Epoch Arrival Time: 1785026466.245953925
[Time shift for this packet: 0.000000000 seconds]
[Time delta from previous captured frame: 1.799384 millisecc]
[Time delta from previous displayed frame: 1.799384 millisecc]
[Time since reference or first frame: 11.923643888 seconds]
Frame Number: 4
Frame Length: 60 bytes (480 bits)
Capture Length: 60 bytes (480 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: ethertype:arp]
Character encoding: ASCII (0)

Figure 7.1 ARP Packet Capture.

7.5. ICMP Packet Analysis

Filter Used

icmp

Observation

The ICMP packet capture displayed Echo Request packets transmitted from Kali Linux and Echo Reply packets returned by the Ubuntu Server.

The successful exchange confirmed that the target machine was reachable and responding correctly to network requests.

Analysis

ICMP packets are commonly used to verify connectivity between hosts. The captured packets demonstrated successful communication between the scanner and the target without any packet loss.

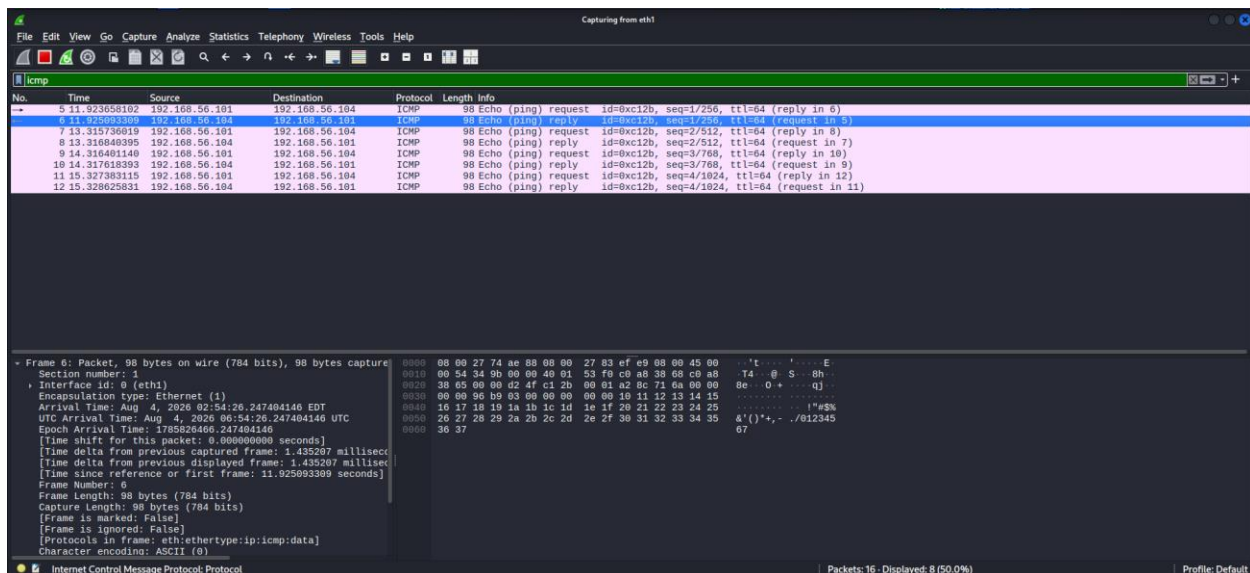


Figure 9.2 ICMP Packet Capture.

7.6. TCP Packet Analysis

Filter Used

tcp

Observation

TCP packets generated during the Nmap port scan were successfully captured. The packet sequence included TCP SYN packets transmitted from Kali Linux and corresponding responses from Ubuntu Server.

For closed ports, the target responded with **RST/ACK** packets, while the open SSH port continued with the TCP connection establishment process.

Analysis

The observed TCP traffic demonstrated how Nmap determines the state of network ports. TCP responses from the target system allowed Nmap to distinguish between open and closed ports.

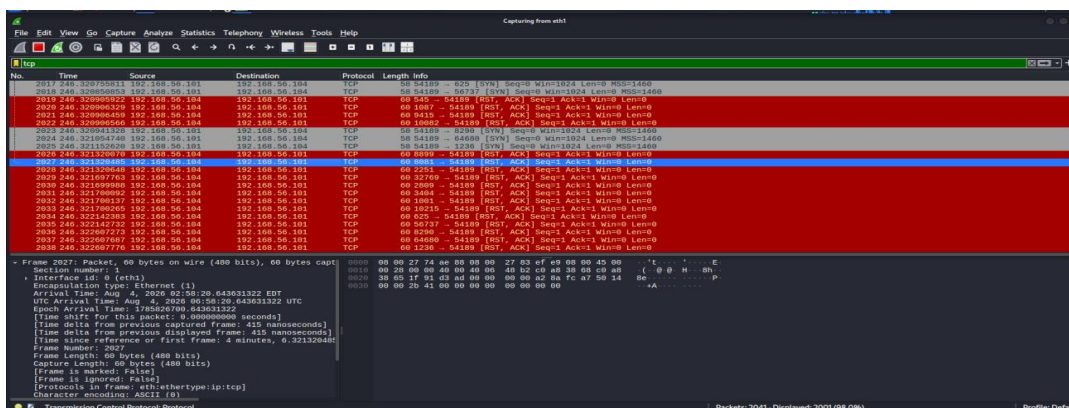


Figure 7.3 TCP Packet Capture.

7.7. SSH Packet Analysis

Filter Used

ssh

Observation

Wireshark captured the SSH communication established between Kali Linux and Ubuntu Server. The capture included protocol negotiation, key exchange, and encrypted packets exchanged after successful session establishment.

Analysis

The captured traffic demonstrated that SSH encrypts communication after authentication and key exchange. Although Wireshark could capture the packets, the encrypted payload could not be interpreted, highlighting the confidentiality provided by the SSH protocol.

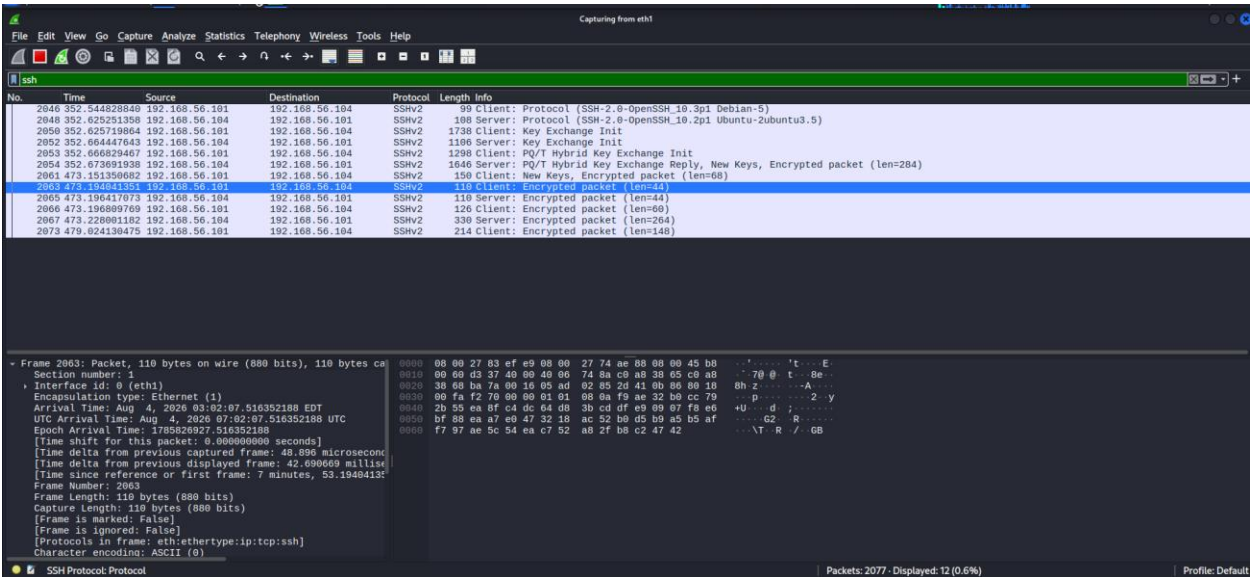


Figure 7.4 SSH Packet Capture.

7.8. Summary of Captured Protocols

Table 7.1 Summary of Captured Network Protocols

Protocol	Purpose	Observation
ARP	IP-to-MAC Resolution	Successfully resolved MAC addresses
ICMP	Connectivity Testing	Echo Request and Echo Reply observed
TCP	Reliable Communication	SYN and RST/ACK packets captured during port scanning
SSH	Secure Remote Access	Encrypted communication successfully observed

7.9. Key Findings

The packet analysis confirmed:

- Successful communication between Kali Linux and Ubuntu Server.
- Correct operation of ARP for local address resolution.
- Reliable ICMP communication with no packet loss.
- TCP behavior during network port scanning.
- Secure and encrypted communication over SSH.

7.10. Chapter Summary

Wireshark successfully captured and analyzed ARP, ICMP, TCP, and SSH traffic generated during the security assessment. The packet captures verified network connectivity, demonstrated protocol behavior, and confirmed secure communication between the scanner and target systems. The observations supported the results obtained during host discovery, port enumeration, and service detection.

Learning Outcome:

Packet analysis provided practical insight into how different network protocols operate during a security assessment. ARP enabled address resolution, ICMP verified connectivity, TCP demonstrated port scanning behavior, and SSH illustrated encrypted communication, reinforcing the theoretical concepts through real network traffic captured in the virtual laboratory.

CHAPTER 8

Risk Analysis

8.1. Introduction

Risk analysis is the process of evaluating the security implications of the findings identified during a network assessment. It helps determine how exposed the target system is to potential threats and assists in prioritizing appropriate security measures. During this assessment, the identified hosts, open ports, running services, and observed network communication were analyzed to determine their potential impact on the security posture of the virtual lab.

8.2. Assessment Summary

The security assessment produced the following key findings:

- Four active hosts were identified within the virtual network.
- Ubuntu Server exposed only one network service (SSH).
- The SSH service was running OpenSSH 10.2p1 using SSH Protocol 2.0.
- The remaining commonly scanned TCP ports were closed.
- Packet analysis confirmed successful and secure communication between Kali Linux and Ubuntu Server.

Overall, the target system presented a limited attack surface with only one exposed administrative service.

8.3. Risk Assessment Table

Table 8.1 Risk Assessment

Finding	Potential Risk	Risk Level	Remarks
SSH Port (22/TCP) Open	May allow unauthorized remote access if weak credentials are used	Medium	Secure configuration and strong authentication are recommended
OpenSSH Service	Outdated software may contain known vulnerabilities if not updated	Low	Current version detected; regular updates should be maintained
SSH Protocol Version 2.0	Secure communication protocol	Low	Recommended protocol version
Closed TCP Ports	Reduced attack surface	Low	Good security practice
Successful Host Discovery	Active devices are visible on the network	Informational	Normal behavior in an internal lab environment

Finding	Potential Risk	Risk Level	Remarks
Packet Analysis	No suspicious or abnormal traffic observed	Informational	Communication appeared legitimate

8.4. Analysis of Findings

Finding 1: SSH Service Exposure

The Ubuntu Server exposed **Port 22**, which is commonly used for secure remote administration.

Although SSH is considered a secure protocol, exposing this service introduces a potential attack vector if weak passwords, default credentials, or outdated software are used.

Risk Level: Medium

Finding 2: Limited Attack Surface

The Nmap scan identified only one open TCP port while the remaining commonly scanned ports were closed.

A smaller number of exposed services reduces the opportunities available for attackers to exploit the system.

Risk Level: Low

Finding 3: Secure Communication

Wireshark analysis confirmed that SSH communication became encrypted after protocol negotiation and key exchange.

This behavior protects sensitive information transmitted between the client and the server.

Risk Level: Low

Finding 4: Network Visibility

Host discovery successfully identified all expected systems within the virtual network.

Since the assessment was performed inside an isolated laboratory environment, the discovered hosts matched the intended network design.

Risk Level: Informational

8.5. Overall Security Assessment

Based on the reconnaissance activities performed during this project, the Ubuntu Server demonstrated a relatively secure configuration.

The assessment identified only one exposed service (SSH), while the remaining commonly scanned ports were closed. The use of SSH Protocol Version 2.0 and encrypted communication contributed positively to the security posture of the system.

No evidence of unnecessary services, suspicious network traffic, or significant security misconfigurations was observed during the assessment.

8.6. Risk Matrix

Table 8.2 Risk Matrix

Risk Level	Description	Findings
High	Immediate security concern requiring urgent attention	None
Medium	Requires appropriate security controls	Open SSH service
Low	Minor security consideration	Service updates, limited attack surface
Informational	General observations without immediate risk	Host discovery and packet analysis

8.7. Chapter Summary

The risk analysis concluded that the virtual lab network maintained a relatively low overall risk profile. The only exposed service was SSH, which is intended for secure remote administration. Appropriate security practices such as strong authentication, software updates, and continuous monitoring would further strengthen the security posture of the system.

Security Assessment Verdict

The network assessment did not identify any critical security vulnerabilities within the defined project scope. The Ubuntu Server maintained a minimal attack surface with only the SSH service exposed. The observed configuration follows basic security best practices for a controlled laboratory environment.

CHAPTER 9

Recommendations

9.1. Introduction

Based on the findings obtained during the network reconnaissance and security assessment, several recommendations are proposed to improve the security posture of the target system. These recommendations focus on reducing potential attack vectors, strengthening authentication mechanisms, and maintaining secure network operations.

9.2. Security Recommendations

1. Use Strong Authentication

Strong and unique passwords should be used for all user accounts to reduce the risk of unauthorized access. Where possible, SSH key-based authentication should be preferred over password-based authentication.

2. Keep the SSH Service Updated

The OpenSSH service should be updated regularly to ensure that the latest security patches and bug fixes are applied. Regular software updates help protect the system against newly discovered vulnerabilities.

3. Disable Root Login via SSH

Direct root login through SSH should be disabled unless it is specifically required. Administrative tasks should be performed using standard user accounts with elevated privileges when necessary.

4. Expose Only Required Services

Only services that are essential for system operation should remain enabled. Unused or unnecessary services should be disabled to reduce the attack surface.

5. Perform Regular Network Scans

Periodic network reconnaissance using tools such as Nmap can help identify newly exposed services, unauthorized devices, or configuration changes before they become security issues.

6. Monitor Network Traffic

Network traffic should be monitored regularly using packet analysis tools such as Wireshark or intrusion detection systems to identify suspicious communication or unusual network behavior.

7. Maintain Secure Network Configuration

Network configurations should be reviewed periodically to ensure that only authorized devices are connected and that unnecessary network exposure is minimized.

9.3. Summary of Recommendations

Table 9.1 Summary of Security Recommendations

Recommendation	Purpose
Use strong passwords or SSH keys	Prevent unauthorized access
Keep OpenSSH updated	Protect against known vulnerabilities
Disable SSH root login	Reduce administrative security risks
Expose only required services	Minimize the attack surface
Perform periodic network scans	Identify new security issues
Monitor network traffic	Detect suspicious activities
Review network configuration	Maintain a secure environment

9.4. Chapter Summary

The recommendations provided in this chapter aim to strengthen the security posture of the assessed network by improving authentication, reducing unnecessary exposure, maintaining updated services, and encouraging continuous monitoring. Implementing these practices can help protect systems against common network-based attacks and improve overall security.

Conclusion

The **Network Reconnaissance and Security Assessment** project successfully demonstrated the process of evaluating the security posture of a virtual laboratory network using industry-standard cybersecurity tools. An isolated environment was created using Oracle VirtualBox with Kali Linux configured as the scanning system and Ubuntu Server as the target machine. This setup provided a safe environment for conducting reconnaissance activities without affecting external networks.

Host discovery was performed using Netdiscover and Nmap, which successfully identified all active devices connected to the Host-Only network. Port and service enumeration revealed that the Ubuntu Server exposed only one open TCP port (22), running the OpenSSH service. Service version detection confirmed the presence of **OpenSSH 10.2p1** using **SSH Protocol 2.0**, while operating system fingerprinting identified the target as a Linux-based system. Although Nmap could not determine an exact operating system match, the results were consistent with the expected Ubuntu Server configuration used in the virtual environment.

Wireshark packet analysis further validated the assessment by capturing and analyzing ARP, ICMP, TCP, and SSH traffic exchanged between the Kali Linux and Ubuntu Server virtual machines. The packet captures demonstrated successful address resolution, network connectivity, TCP communication during port scanning, and encrypted SSH communication after authentication. These observations reinforced the practical understanding of how different network protocols function during a security assessment.

Based on the collected evidence, the overall security posture of the target system was found to be satisfactory within the defined scope of the project. The Ubuntu Server maintained a limited attack surface with only the SSH service exposed, while the remaining commonly scanned ports were closed. No unnecessary services or suspicious network activity were observed during the assessment.

The recommendations provided in this report, including the use of strong authentication, regular software updates, continuous network monitoring, and periodic security assessments, can further strengthen the security of the target system. Overall, this project enhanced practical knowledge of network reconnaissance, packet analysis, and security assessment methodologies while providing valuable hands-on experience with professional cybersecurity tools.

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